

NEERAJ KUMAR

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EDUCATION

Guru Gobind Singh Indraprastha University, New Delhi
Bachelor's Degree - Automation & Robotics - CGPA - 7.00

Nov 2022 – Jul 2026
New Delhi

TECHNICAL SKILLS

Languages: C++17/20, C, Python, JavaScript

Technologies/Frameworks: React, Node.js, Express, PostgreSQL, MongoDB, Redis, WebSockets, Firebase, Docker, Linux/UNIX environments, REST APIs, TCP/IP

Tools: Git, Nginx, GitHub Actions, Vercel, Postman, VS Code

Core Competencies: Data Structures and Algorithms, Object-Oriented Programming, Operating Systems, System Design, Distributed Systems

EXPERIENCE

Outlier - AI Code Reviewer & Data Annotator
Freelance

Mar 2026 – Apr 2026
Remote

- Reviewed and evaluated AI-generated code and technical responses for correctness, logic, edge cases, and adherence to task requirements.
- Annotated and ranked model outputs, identifying logical errors, inefficient implementations, and inconsistencies to improve code-generation quality.
- Applied structured evaluation criteria across programming and technical tasks, providing detailed feedback and corrected solutions.

PROJECTS

Discord-Vault | [GitHub](#) | Python, SHA256

Jul 2026 – Sep 2026

- Developed a secure Discord-based file vault for storing and retrieving encrypted files through a bot interface.
- Implemented **SHA-256** based integrity verification to detect file tampering and ensure data consistency.
- Implemented lossless PNG compression and decompression, reducing image storage overhead while preserving pixel-perfect data integrity.
- Automated file upload, retrieval, and vault management through Discord commands, eliminating the need for a dedicated frontend.

ClipIt | [GitHub](#) | Python, Next.js, Docker, FFmpeg

Dec 2025 – Jan 2026

- Designed a **concurrent video processing backend**, handling multiple clipping requests with collision-free file management.
- Implemented **thread-safe ID generation**, ensuring deterministic output under parallel workloads.
- Integrated **FFmpeg** pipeline for efficient video slicing, reducing processing overhead.
- Deployed containerized service using **Docker**, enabling reproducible builds and scalable execution.

Options Pricing Simulator | [GitHub](#) | Python, SciPy, NumPy

Feb 2026 – Mar 2026

- Developed a **Monte Carlo** simulation engine to price **European Options** with configurable simulation runs.
- Validated results against **Black-Scholes** analytical model, achieving **<0.03** pricing deviation at **10,000** simulations.
- Computed option sensitivities : Delta (Δ), Vega (ν), Gamma (Γ), and Theta (Θ) - to support quantitative risk analysis.

RedisCLI | [GitHub](#) | C++17, POSIX Sockets, RESP2

Mar 2026 – Apr 2026

- Built a Redis command-line client from scratch in **C++17**, implementing the full **RESP2** protocol serializer and parser.
- Implemented event-driven **Pub/Sub** mode using **poll()** to multiplex stdin and Redis socket simultaneously - streaming published messages while keeping terminal input responsive via GNU Readline callback API.

CERTIFICATIONS

- Postman API Fundamentals Student Expert** - [Postman](#)
- Career Essentials in GitHub Professional Certificate** - [LinkedIn Learning](#)
- Software Engineering Job Simulation (JPMorgan Chase & Co.)** - [Forge](#)

ACHIEVEMENTS & EXTRACURRICULARS

- Achieved Max Rating of **1614** on [LeetCode](#); solved **300+** algorithmic problems on LeetCode in C++.
- Achieved Global Rating of **123** on CodeChefs and max rating of **1159** on [Codeforces](#).
- Represented the college Volleyball team in inter-collegiate tournaments and secured **3rd place**.